Classy.

Technical documentation

Classy is a web app that gives GUtech students their own class timetable, built from the university's EduWave data. This document explains how it is built, hosted and kept up to date. Endpoint details are in the separate Classy API Documentation.

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01

Overview.

Classy shows each GUtech student their real timetable, class by class and date by date, and is live at schedule.tryzarqa.com. Students sign in with their university ID, pick their major, year, courses and sections once, and then see today's classes, the week, and any changes.

STUDENTS

See their timetable, get a "Changed" badge when a class moves, add their own slots, write private or public notes on classes, pick colour themes, share their week as a picture, and use groups to find free time with friends or plan a shared car ride.

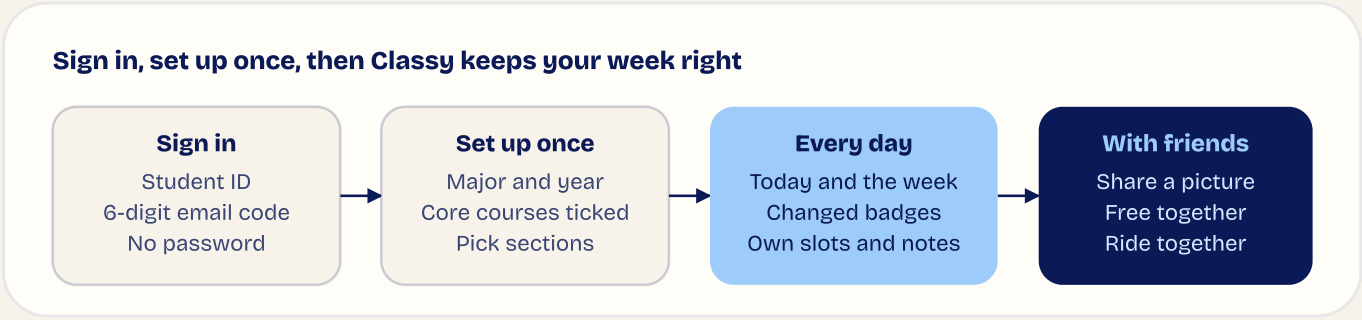
ADMINS

Run Classy from a dashboard: check and apply timetable updates, set default courses, moderate public notes and manage accounts.

02

Features.

Classy has 15 features for students and 12 for staff, all live today.



A student's path through Classy, in four stages.

For students

FEATURE	WHAT IT DOES
Sign in with your ID	A 6-digit code to the GUtech student email; no password, signed in for 60 days
Guided setup	Pick major and year; core courses are already ticked, then life skills, electives and sections
Get my courses from GUtech	Optional. One press fills in courses and sections, if GUtech also offers a student-courses API
Today	The class on now with the minutes left, a countdown to the next one, and the whole next day on free days
Week and day views	Numbered teaching weeks (Week 1, Week 2), every class on its real date, long classes at full length

FEATURE	WHAT IT DOES
Changed badge	Shows when a class gets a new time, room or teacher, until the student has seen it
Clashes	Overlapping classes are flagged, never blocked, for students with a clash waiver
Own slots	Gym, study or work, every week or on one date, next to the classes
Class notes	On a course or one class date; private, or shared with classmates, who can report them
Colours	Preset themes, or the student's own main colour, accent and background
Change year	From the menu; core courses switch to the new year's
Share as a picture	The week or day as a square or story-sized picture in the student's colours
Free together	A group sees when everyone is free each week, and when all but one or two are
Ride together	A car group sees when to leave home and uni each day, and who waits how long
Rate Classy	A rating pop-up when staff ask for feedback

For staff

FEATURE	WHAT IT DOES
Timetable upload	Excel, CSV or pasted rows, with a preview of every change and the students affected
EduWave sync	The Classy Sync extension fetches the timetable, every morning if wanted
Connections	Any JSON timetable or student-courses API, set up without code
Edit a section	Correct a teacher, time or room, or add a class
Default courses	Per major, family of majors and study year, by drag and drop
Majors and electives	Families of majors, and extra courses a major may take
Semesters and weeks	Create, publish and delete semesters; set week 1 and the last teaching week
Notes	Every public note in one place, reported ones flagged; hide or delete
Feedback	Send a "Rate Classy" question; see stars and anonymous comments
Dashboard	Students active now and today, setup progress, what needs attention, recent uploads
Accounts	Admin accounts; new admins set their own password
Owner tools	See individual students, reset their setup, make owners, delete all data (three confirmations)

03

Timetable data.

Today and the plan.

Today: reading EduWave

The Classy Sync browser extension opens EduWave's Advisor Plan page in an owner's signed-in browser and asks for each course's sections, the same request the page's own "Sections" button sends. It reads one course a second, about 4 minutes for 217 courses, and staff apply the result. This works, but it has limits:

- It depends on one person's EduWave session staying signed in.
- Someone has to run it or leave the browser open for the morning sync.

The plan: an official API

A read-only feed from GUtech would make Classy exact and hands-off. This is what Classy needs:

FIELD	NEEDED	EXAMPLE
Course code	Yes	CS 1 107
Course name	Yes	Data Structures
Section number	Yes	3
Instructor	Yes	Nabil Sahli
Class date	Yes	2026-10-18
Start and end time	Yes	09:45–11:15
Room	Yes	GU1 240
Programme (major) and study year	Helpful	CS, Year 1
Course type and credits	Helpful	Core, 5
Last changed time	Helpful	2026-10-17 08:12

Access could be an API key or a service account, read-only, ideally with a way to ask for "changes since" a time. No student records are needed: students still sign in and pick their own sections.

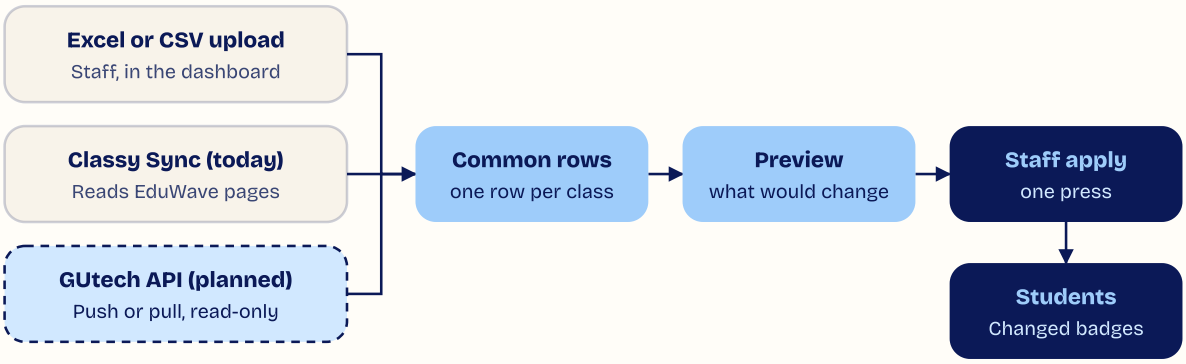
OUR OWN DOMAIN

Classy runs at schedule.tryzarqa.com for now. Once GUtech's API is connected, we will buy Classy its own domain and move to it. The API stays the same; only the address changes.

Why Classy is ready for any API

Every source already goes through one import pipeline. Each source is turned into the same simple rows, one per class, and everything after that is shared and tested.

Any source, including a GUtech API, feeds the same import steps



An official API would be one more source on the left; the steps from common rows to students stay as they are.

ANY JSON API

Set up in the dashboard: owners enter its address, a sign-in key and which field holds what. No code changes, and a Test button shows what Classy reads first.

TIMETABLE, AND OPTIONALLY STUDENT COURSES

Classy fetches the timetable on demand or every morning. An optional student-courses API (with `{studentId}` in its address) gives students a "Get my courses from GUtech" button.

OR GUTECH SENDS IT

To `POST /api/sync/timetable` and `POST /api/sync/enrollments`, with a token.

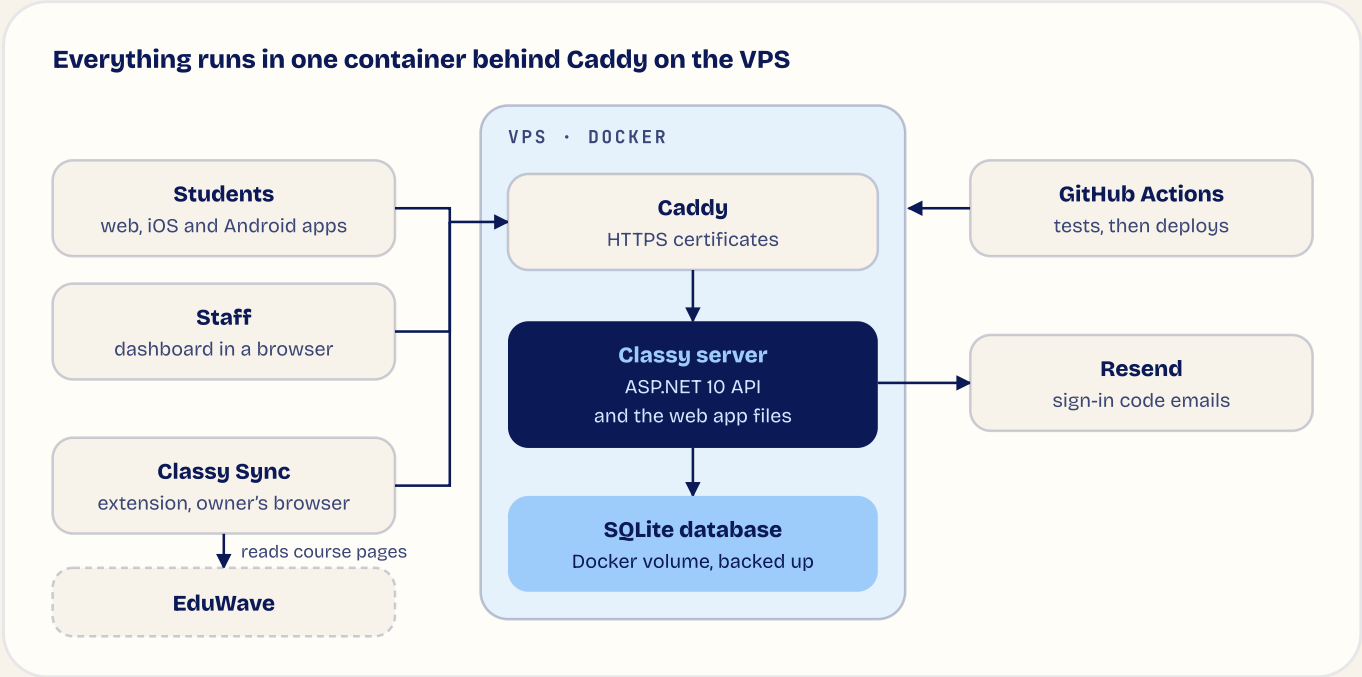
SAFE BY DEFAULT

Staff preview every change before students see it.

04

Architecture.

Students, staff and the sync extension all reach one server over HTTPS. That server keeps everything in one database file and only talks to an email service.



The web app and the API come from the same container, so there is one address and no cross-site setup. The VPS never contacts EduWave; only the extension in an owner's browser reads it.

05

Technology.

Classy is one .NET server and one React web app, shipped together in a single Docker container.

PART	TECHNOLOGY
Server	C# on .NET 10 (ASP.NET Core)
Database	SQLite with Entity Framework Core
Web app	React 19 and TypeScript, built with Vite; Tailwind CSS
Mobile apps	Capacitor shells around the web app
Timetable sync	Chrome/Brave extension
Email	Resend (sign-in codes only)
Hosting	Linux VPS, Docker, Caddy for HTTPS
Tests and deploys	xUnit, GitHub Actions

06

Data model.

All data is in one SQLite file (`schedule.db`) with 24 tables. A semester holds courses, a course holds sections, and a section holds its dated classes. Students link to that through their picks.

GROUP	TABLES	WHAT THEY HOLD
Timetable	Semester, Major, Course, CourseMajor, MajorElective, CourseDefault, Section, Meeting	Semesters, majors, courses, sections and every dated class
Data sources	UploadLog, PendingSync, SyncToken, Connection, Enrollment	Uploads and syncs waiting to be applied, connected APIs, courses GUtech sends
People	Student, AdminUser, LoginCode	Students (ID, email, major, year), staff accounts, pending sign-in codes
Student data	Pick, CustomSlot, ClassNote, NoteReport, FeedbackPrompt, FeedbackResponse, StudyGroup, GroupMember	Course picks, own slots, notes and reports, feedback, groups and their members

Deleting a student removes their picks, slots, notes, reports, feedback and group memberships with them. Times are stored in UTC. Schema changes go through EF Core migrations, which run when the server starts.

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How it works.

Sign-in

Students have no password. They enter their university ID, Classy emails a 6-digit code to their `@student.gutech.edu.om` address, and the code starts a session cookie that lasts 60 days and renews while used. Staff sign in with a username and a password of at least 12 characters, and new staff accounts must change their temporary password first. Every request checks the account and its role in the database again.

Student setup

Done once per semester, in this order:

1. Pick a major. Classy suggests the study year from the ID's intake year (24-xxxx), counting the foundation year.
2. Confirm the study year (Foundation, or 1 to 5).
3. Core courses for that major and year are already ticked from the defaults staff set; the student can change them.
4. Add life skills and electives.

5. Pick a section for each course. Clashes are shown but allowed, because some students have clash waivers.

Students can change their year later from the menu; their core courses then switch to the new year's.

CHANGE BADGES

When a section's teacher, times or rooms change, Classy records when and what changed. Students who picked it see a "Changed" badge until they open the class. If a section is removed, its students are asked to pick another.

NOTES AND MODERATION

Students write notes on a course or on one class date. Private notes are seen only by their author. Public notes are seen by classmates taking the course (for a class date, those in the same section), with the author's university ID. Students can report a note, and staff can hide or delete public notes; private notes never appear in the dashboard. Posting is limited to 30 notes an hour.

GROUPS AND RIDES

Students make a group and send its invite link; others join only through it, under a name they choose. Classy works out on the server when everyone is free that week, or, for a car group, when to leave home and uni each day and who waits how long.

SHARING A PICTURE

The Share button on the week and day views draws the timetable as a picture in the student's colours, on the phone itself, and opens the phone's share sheet.

08

Hosting and deployment.

Classy runs as one Docker container on a Linux VPS, behind the Caddy web server, which provides HTTPS for schedule.tryzarqa.com. DNS is managed in Cloudflare.

CONTAINER

One Docker image holds the web app and the .NET server, running as a non-root user.

DEPLOYS

Every push to `main` runs the tests on GitHub Actions; if they pass, the server is updated over SSH.

BACKUPS

The database is backed up before every deploy, and the newest 14 copies are kept.

SECRETS

Secrets stay in GitHub and a settings file on the server, never in the code.

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Testing.

The server has 133 automated tests covering sign-in, roles, timetable imports, student features, the dashboard and outside APIs. All must pass, and the web app must build cleanly, before any deploy.

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Security and privacy.

How accounts, sign-in and data are protected.

Stored per student: university ID, student email, major, study year, course and section picks, their own slots and notes, feedback answers, and when they were last active. No names or passwords.

Sign-in codes expire after 10 minutes, allow 5 wrong tries, and are stored salted and hashed. Staff passwords are hashed, and 5 wrong passwords lock the account for a while.

Sessions use an HttpOnly, SameSite cookie, so page scripts can't read it. Roles are checked against the database on every request, so a removed account loses access at once.

Rate limits per university ID: one sign-in code a minute and four every 15 minutes, and 30 notes an hour. A much higher cap per network only stops floods, since everyone on campus Wi-Fi shares one address.

EduWave: Classy never asks for or stores an EduWave password, and the server never signs in to EduWave.

Transport and storage: HTTPS only. The database lives on the VPS in a Docker volume, backed up before every deploy.